Intermediate Computer Science

A self-study companion · MathJax · SVG diagrams · Flexbox layouts

12 UNIT BOOKLETS | EMBEDDED MATHJAX 3 | HAND-DRAWN-STYLE SVG | PRINT-FIRST A4

How these notes were built. Each booklet follows the official intermediate syllabus units (digital systems through digital entrepreneurship). Every chapter is a single HTML file: Flexbox cards for comparisons, inline SVG for gates/stacks/networks, and MathJax for the notation an examiner expects. Chrome prints the same file to the PDF you are holding.

Study rule: read one section, redraw the figure from memory, then attempt the self-check before you turn the page.

Companion to the Class XI / XII Computer Science textbooks · not a replacement for practicals (LogiSim, Access, Figma, Python labs)

Class XI — six units

Year 1 pathway

Unit	Booklet	You should leave able to…
1	Computer Systems	Write $Y = A \cdot B + C$, simplify on a K-map, contrast Waterfall/Agile (Al-Noor), map OSI ↔ TCP/IP
2	Computational Thinking	Decompose / pattern / abstract; trace bubble & selection sort; choose linear vs binary search
3	Programming Fundamentals	Use `input` / `int`, `if/elif`, `for` / `while`, `math`, and name the three bug families
4	Data and Analysis	Draw the library ER (Book–Borrow–Member), name keys, build Access tables/forms/queries
5	Applications & Impacts	Sketch the IoT loop; separate AI / IoT / analytics; judge a source; name assistive tools
6	Digital Literacy	Write a research question, pick methods, search with `filetype:` / `site:`, present without plagiarising

Class XII — six units

Year 2 pathway

Unit	Booklet	You should leave able to…
1	Human–Computer Interaction	Apply contrast and inclusive navigation; wireframe → Figma prototype; evaluate with SUS / task success
2	Algorithms & Data Structures	Trace for correctness; quote $O(1)/O(\log n)/O(n)/O(n^2)$; PUSH/POP vs enqueue/dequeue; tree vs graph
3	Python Structures & Files	Choose list/tuple/set/dict; write functions with scope; use `with open` and `try/except`
4	Data Analysis with Python	Connect `sqlite3`, clean a DataFrame, pick bar/hist/scatter/box, compute $\bar{x}$, median, σ
5	Neural Nets & Security	Separate ML / NN / DL; label layers; name phishing + 2FA; distinguish equal access from equity
6	Digital Entrepreneurship	Walk problem → prototype → MVP → beachhead; test the riskiest assumption first

Notation you will keep meeting

$$Y = A \cdot B + \overline{C}, \quad MID = \left\lfloor \frac{BEG + END}{2} \right\rfloor, \quad T(n) = O(\log n), \quad \bar{x} = \frac{1}{n}\sum x_i, \quad z = w \cdot x + b$$

FLEXBOX IN THESE NOTES

Comparison cards, badge rows, two-column explainers, and the SDLC/IoT/sprint strips are all `display:flex` so a printed page and a browser window share one layout language.

SVG, NOT SCREENSHOTS

Gates, waves, OSI stacks, stacks/queues, and neural nets are line diagrams you can re-sketch in an exam hall. If you cannot redraw it, you do not own it yet.

MATHJAX

Boolean algebra, Big-O, statistics, and the neuron sum are typeset with MathJax 3 (SVG output) so PDF and HTML stay sharp when zoomed.

Suggested weekly rhythm

1. One booklet section a sitting (not a whole unit in one night).
2. Redraw the figure; write the formula without the booklet.
3. Do the self-check; only then peek at the answers.
4. Transfer the idea into the lab tool of that unit (LogiSim, Python, Access, Figma, a paper MVP).